

## Evaluation

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**Please provide your assessment of the work according to the criteria below.**

### **Makes an original and significant contribution to knowledge or understanding in its field(s)**

The thesis at hand by Mr Sobhan Chatterjee is an excellent contribution to the field of Machine Learning (ML) for Cyber-Physical Systems (CPSs). With his work, he has clearly extended the state of the art in the field by developing methods for compositional ML, which lends itself better to verification and to gaining insights into the behavior of the ML components of a system. Along with his thesis, he has published three papers in reputable venues and has one more under submission. The topics addressed are timely and highly relevant. ML techniques are currently seeing unprecedented growth. Therefore, methods to better apply them in the context of CPSs are urgently needed. The addressed challenges of safety verification, explainability, and predictable timing are critical, and his contributions are an important step toward wider acceptance of ML.

### **Meets internationally recognised standards for such work**

The thesis at hand clearly meets internationally recognized standards and is an excellent example for thorough PhD research. The core contributions have been accepted in high-quality peer-reviewed international venues, which supports my assessment that the international standards are met by the thesis.

### **Demonstrates knowledge of the literature relevant to the subject and the field(s) to which the subject belongs, and demonstrates the ability to exercise critical and analytical judgement of that literature**

Mr. Chatterjee has thoroughly reviewed the body of literature relevant to his thesis. He has clearly identified existing research gaps and filled them with his research. All his contributions are put in context with the scientific state of the art and he nicely illustrates how he is extending upon existing works by putting his contributions in relation with prior works.

### **Is satisfactory in its methodology, in the quality and coherence of its expression, and in its scholarly presentation and format**

The methodology used for the thesis corresponds to the international scientific standards. The problems addressed are well motivated. All hypotheses are extensively built up and discussed in context with the state of the art. The developed methodologies are implemented and thoroughly evaluated with experiments. The quantitative evaluation is compared with results from state-of-the-art works and illustrates the novelty of the developed solutions. The thesis is very well written and shows a high quality and coherence. All contributions surround the proposed framework and nicely fit together to support the motivation of the thesis. As the core chapters of the thesis have been published in reputable conferences and journals, the scholarly presentation is thorough and the format is fine.

### **Additional comments**

See minor recommended revisions

### **Questions for the oral examination**

Chapter 3:

- Why is the compositional approach not commonly used if it is beneficial?
- Please comment on the <5% classification error. Can you put that into context?

Chapter 4:

- How far could the decomposition into separate models be performed automatically? Currently, human reasoning is used to separate functionalities (such as splitting the lane change decision into steps).

Chapter 5:

- Could the architecture of the compositional model be obtained from an optimization approach? One might have missed the best architecture in terms of the number of stages and cluster models.

## Recommended revisions, if any

Recommended minor revisions:

The thesis is excellently written and shows a high scientific quality. However, some minor formatting issues do not match the excellent scientific contributions:

- General: The figures in the thesis differ in their visual language. Colors, fonts, and sizes are not consistent across the thesis. Also, the label fonts inside figures vary strongly w.r.t. their size.
- General: The mathematical operator for multiplication is typed inconsistently: it should be typed with a “\cdot” command in LaTeX. However, in, e.g., equation 1.1 it is written as “\*” and in equation 1.17 as “x”.
- General: In equations and algorithms, the font used for indices and variables changes. In chapter 3, it is italic (without use of \textsl), and in chapter 4, it is using \mathrm.
- Page 2, 1.2.2., sentence 2: “respond”->“responds”
- Page 8, 1.3.2.1, equations 1.2 to 1.6 are labeled, but in the preceding text, they are only listed and referenced “in the same order”. Directly referring to labels would be easier to follow.
- Page 31: Table 2.1 extends beyond the text area.
- Page 36: Text before Figure 2.3: Reference to Figure 3.2 -> should be Figure 2.3.
- Page 44: Figure 3.1 caption has no “.” at the end. Figure 3.2 is very low resolution.
- Page 50: Equation 3.2: Italic “Model” looks strange, use \textsl in equations.
- Page 89: “.” After figure caption missing in Fig. 4.4 – 4.7.
- Page 92: first text line: “Given” wrong capitalization
- Page 106: Table 5.1 extends beyond page margin, more tables like this in chapter 5.